

Ros2 day_1 hw4 Report

로봇 21기 예비단원 최동현

```
^C[INFO] [1789460362.034766324] [rclcpp]: signal_handler(SIGINT/SIGTERM)
choi@choi-15Z90Q-GRLGL:~/ros2_ws$ cd ~/ros2_ws
colcon build --packages-select service_pkg
source install/setup.bash
ros2 run service_pkg add_server
Starting >>> service_pkg
Finished <<< service_pkg [2.37s]

Summary: 1 package finished [2.45s]
[INFO] [1789460373.899275224] [calc_server]: 서비스 준비 완료: add / minus / multiply / divide
[INFO] [1789460378.522673086] [calc_server]: 요청: 12 + 4 = 16
[INFO] [1789460379.480915219] [calc_server]: 요청: 12 - 4 = 8
[INFO] [1789460380.179824828] [calc_server]: 요청: 12 * 4 = 48
[INFO] [1789460380.884114490] [calc_server]: 요청: 12 / 4 = 3
[WARN] [1789460381.586124032] [calc_server]: 0으로 나눌 수 없습니다 (b=0)
□
```

```
choi@choi-15Z90Q-GRLGL:~/ros2_ws$ ros2 service call /add_two_ints example_interfaces/srv/AddTwoInts
s "{a: 12, b: 4}" # → 16
ros2 service call /minus_two_ints example_interfaces/srv/AddTwoInts "{a: 12, b: 4}" # → 8
ros2 service call /multiply_two_ints example_interfaces/srv/AddTwoInts "{a: 12, b: 4}" # → 48
ros2 service call /divide_two_ints example_interfaces/srv/AddTwoInts "{a: 12, b: 4}" # → 3

# 0으로 나누기 테스트 (서버 안 죽고 경고만 떼야 정상)
ros2 service call /divide_two_ints example_interfaces/srv/AddTwoInts "{a: 12, b: 0}" # → sum=0, 서버에 WARN
waiting for service to become available...
requester: making request: example_interfaces.srv.AddTwoInts_Request(a=12, b=4)

response:
example_interfaces.srv.AddTwoInts_Response(sum=16)

waiting for service to become available...
requester: making request: example_interfaces.srv.AddTwoInts_Request(a=12, b=4)

response:
example_interfaces.srv.AddTwoInts_Response(sum=8)

waiting for service to become available...
requester: making request: example_interfaces.srv.AddTwoInts_Request(a=12, b=4)

response:
example_interfaces.srv.AddTwoInts_Response(sum=48)

waiting for service to become available...
requester: making request: example_interfaces.srv.AddTwoInts_Request(a=12, b=4)

response:
example_interfaces.srv.AddTwoInts_Response(sum=3)

waiting for service to become available...
requester: making request: example_interfaces.srv.AddTwoInts_Request(a=12, b=0)

response:
example_interfaces.srv.AddTwoInts_Response(sum=0)
choi@choi-15Z90Q-GRLGL:~/ros2_ws$ □
```

<실행결과>

service통신을 이용하여서 노드끼리 보내면 받고를 진행하였다. 보낸 데이터는 숫자와 사칙연산종류였다.
더하기, 빼기, 곱하기, 나누기를 구현하였다.

```
rclcpp::Service<AddTwoInts>::SharedPtr srv_add_;  
rclcpp::Service<AddTwoInts>::SharedPtr srv_minus_;  
rclcpp::Service<AddTwoInts>::SharedPtr srv_multiply_;  
rclcpp::Service<AddTwoInts>::SharedPtr srv_divide_;
```

사칙연산 부호는 이렇게 다 따로 받았다.